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ELIMINATE DEAD CODE

Aim:

To develop a C program that detects and removes dead code in Three Address Code (TAC), keeping only statements needed to produce the final output.

1. Start the program.
2. Read the Three Address Code statements until END is entered.
3. Separate each statement into its left-hand side (LHS) and right-hand side (RHS).
4. Examine the statements in reverse order to find which results are needed.
5. Mark statements whose values are used by other required statements.
6. Keep assignments to the final output variables x and y.
7. Identify and discard statements whose results are never used.
8. Print the remaining optimized Three Address Code.
9. Stop the program.

Pseudocode:

```
START
READ TAC statements until END
FOR each statement
   Separate LHS and RHS
END FOR
```

```
FOR each statement in reverse order
    IF statement has no LHS
        Mark it as live
    ELSE IF its result is required
        Mark it as live
    END IF
    IF LHS is x or y
        Mark it as live
    END IF
    IF statement is live
        Mark earlier definitions of its RHS variables
    END IF
END FOR
PRINT all live statements
STOP
```

Output:

Output

```
Enter Three Address Code statements.
Enter END to finish:
t1 = a + b
t2 = c * d
t3 = t1 + 10
x = t3
y = 20
END

Optimized Three Address Code:
t1 = a + b
t3 = t1 + 10
x = t3
y = 20
```

Result:

Thus the C program successfully eliminates dead code from Three Address code by removing the unused statement "t2=c*d" and displaying the optimized code